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4NF (4th Normal Form)

Defn A relation R is in 4NF if and only if the following condn are satisfied →

1. If R is already in 3NF or BCNF.
2. If it contains no MVD (multivalued dependency).

MVD (Multi-valued dependency)

It is the dependency where one attribute value is potentially a 'multi-valued fact' about another.

★ (a) There must be 3 or more attributes.
(b) Attributes must be independent of each other.

Person (P)	Mobile (M)	Food (F)
P_1	M_1 M_2	F_1 F_2
P_2	M_3	F_2

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	Person (P)	Mobile (M)	Food (F)
$t_1 \rightarrow$	P_1	M_1	F_1
$t_2 \rightarrow$	P_1	M_2	F_2
$t_3 \rightarrow$	P_1	M_1	F_2
$t_4 \rightarrow$	P_1	M_2	F_1
	P_2	M_3	F_2

(1) $P \twoheadrightarrow M$

(2) $P \twoheadrightarrow F$

mobile & food are independent of each other.

M.V.D. $\alpha \twoheadrightarrow \beta$ if any legal relation $r(R)$ for all pairs of tuples t_1 and t_2 in r such that $t_1[\alpha] = t_2[\alpha]$, ~~then there~~ exists tuples t_3 and t_4 in r such that

(a) $t_3[\alpha] = t_4[\alpha] = t_1[\alpha] = t_2[\alpha]$

(b) $t_3[\beta] = t_1[\beta]$ ($\because$ M1 = M1)

(c) $t_4[\beta] = t_2[\beta]$ ($\because$ M2 = M2)

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Q3 Consider the relation $R = (A, B, C, D, E, F, G, H, I)$

$$F = \{ \begin{aligned} AB &\rightarrow C \\ A &\rightarrow DE \\ B &\rightarrow F \\ F &\rightarrow GH \\ D &\rightarrow IJ \end{aligned} \}$$

What is the key of R? and also decompose R into 2NF, then 3NF relations.

Ans:

$R_1 = \{ADEIJ\}$ $R_2 = \{BFGH\}$ $R_3 = \{ABC\}$

$R_{11} = \{DIJ\}$ $R_{21} = \{FGH\}$

$R_{12} = \{ADE\}$ $R_{22} = \{BF\}$

Final set of relations in 3NF are $(R_{11}, R_{12}, R_{21}, R_{22}, R_3)$

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